

need systems that can interpret data, anticipate outcomes, and adapt in real time. Intelligent automation begins where systems move from observation to understanding, and from execution to informed decision-making.

Intelligent Automation Is About Resilience First

There is a tendency to view intelligent automation primarily through the lens of efficiency. In reality, its greatest value lies in resilience.

AI-led automation enables factories to move from reactive operations to predictive and adaptive ones. Equipment health can be assessed continuously. Energy consumption can be optimised dynamically. Production lines can be balanced based on real-time constraints rather than fixed assumptions made months earlier.

Equally important, this shift is not about removing people from the equation. The most effective implementations are those where humans and machines work in deliberate synergy—machines providing speed, precision, and predictive insight, while people apply judgment, experience, and contextual understanding.

This matters deeply in India, where manufacturing spans a wide spectrum—from advanced greenfield facilities to MSMEs operating legacy equipment. Intelligent automation allows intelligence to be layered gradually, enabling modernization without disruptive overhauls or excessive capital pressure.

Digital Twins Are Becoming Practical Decision Tools

Digital twins were once viewed largely as design-stage tools. That perception is changing quickly.

Today, digital twins are becoming operational assets. They help manufacturers answer practical questions before committing resources. What happens if a production line is reconfigured? How does energy usage change when throughput increases? What are the downstream effects of a process modification?

By simulating scenarios virtually,

manufacturers reduce risk on the shop floor. Decisions become more deliberate and less experimental.

From Industry 4.0 to Industry 5.0—Quietly and Practically

As automation advances, the narrative is shifting toward Industry 5.0, with a renewed focus on people.

Machines excel at speed, precision, and consistency. Humans excel at judgment, creativity, and contextual understanding. The future of manufacturing lies in combining these strengths, not choosing between them.

For India, this approach is particularly important. Automation should be seen as a pathway to skill evolution and productivity growth, not as a threat to employment.

The Often-Overlooked Role of the Digital Backbone

Behind every smart or autonomous factory lies a robust digital backbone. Industrial IoT provides the sensing layer. Digital twins offer contextual understanding. Edge computing and analytics enable real-time decisions at the shop-floor level, without latency or dependency on centralised systems.

As factories scale toward higher levels of intelligence, software platforms are emerging as the central nervous system of modern manufacturing. Beyond individual machines, plant-wide supervisory and automation software enables seamless connectivity, real-time monitoring, and data-driven operational control across lines and facilities. At Delta Electronics India, this approach is reflected in platforms such as VTScada and the DIAEP+ Equipment Automation Program, which help unify equipment automation, process visibility, and performance analytics within a single operational framework.

Together, these elements form a digital thread that connects design intent to operational execution across the value chain. This continuity is essential for achieving traceability, compliance, quality consistency, and energy efficiency, as global customers demand

higher transparency and accountability.

From the Shop Floor to the Top Floor

One of the most important shifts enabled by intelligent automation is the closing of the gap between the shop floor and the top floor.

Operational insights around productivity, quality, energy consumption, and asset health can now flow directly into management decision-making. At Delta Electronics India, automation solutions span from shop-floor operations to enterprise-level analytics, enabling organizations to align strategic intent with consistent execution on the shop floor and informed decision-making at the top floor.

Sustainability and Skills as Core Manufacturing Metrics

Sustainability is no longer a parallel discussion. Energy efficiency, emissions reduction, and resource optimisation are now central to manufacturing competitiveness. When energy data, automation control, and analytics come together, productivity and sustainability stop competing with each other.

No automation strategy succeeds without people who understand and trust the technology. As factories become more intelligent, demand for digital and analytical skills will grow sharply. Upskilling must be embedded into automation roadmaps from the start, particularly for MSMEs.

Looking Ahead

India's ambition to emerge as a global manufacturing powerhouse will not be achieved through capacity expansion alone. It will be realized by embedding intelligence across factories—allowing them to sense, learn, and adapt continuously.

In a world defined by disruption, the factories that succeed will not be those that simply produce more. They will be the ones that respond faster, recover better, and improve continuously. Intelligent automation is no longer optional—it is the foundation of India's industrial competitiveness in the coming decade.